

Martingale Flow Matching

Arbitrage-free generative models of asset-price dynamics via constrained stochastic bridges — with convergence theory and a pricing-error certificate.

Status proposal v2.0 · theorems developed & adversarially verified Date August 2026

Domain generative modeling × mathematical finance

Verification literature-audited, see §4

ABSTRACT

Generative models of asset-price paths power pricing engines, hedging simulators, and stress tests — yet almost none of them respect the single non-negotiable law of derivatives markets: under the pricing measure, discounted prices must be *martingales* whose marginal distributions match the ones the option market itself implies. We propose **Martingale Flow Matching (MFM)**, the first scalable learning algorithm that constructs neural diffusion bridges constrained to be martingales with option-implied marginals. The scheme alternates two Kullback–Leibler projections — a marginal-matching step in the style of flow matching, and a martingale-restoration step implemented as an adversarial game against learned trading strategies — and we prove it converges to the entropic martingale optimal transport solution. In the single-asset case, we further prove a *pricing-error certificate*: the model's error on path-dependent payoffs is controlled by two measurable training quantities. The mathematical foundations (Bass martingales, martingale Schrödinger bridges) were established by the stochastic-analysis community within the last year; no learning algorithm exists. This proposal supplies it.

§1 · THE IDEA IN PLAIN LANGUAGE

Why this matters, without the mathematics

When a bank prices an exotic option, runs a stress test, or trains a hedging engine, it needs to simulate thousands of plausible futures for asset prices. Modern practice increasingly uses deep generative models — the same technology

behind image generators — to produce these futures. The problem is that financial markets impose a law that images do not: *you cannot generate scenarios that contain free money*.

Two things make a set of simulated price paths financially legitimate. First, no trading strategy should be able to profit from the simulation on average without taking risk — this is the *no-arbitrage* principle, and mathematically it says simulated prices must be a martingale: a process whose best forecast of tomorrow is today. Second, the simulation must agree with what real option markets are already telling us: the prices of traded options pin down the full probability distribution of the asset at each option expiry, and a legitimate simulator must reproduce those distributions exactly, not approximately.

Today's neural market simulators satisfy neither constraint, or satisfy them only with soft penalties that leave residual violations. A hedging engine trained on such a simulator learns, in part, to harvest fictitious arbitrage — money that exists only inside the simulation. A risk system built on it quietly misprices the tails. As generative models become financial infrastructure, this gap becomes a structural flaw in that infrastructure.

Our proposal closes the gap. We show how to train a generative model that is a martingale *by construction* and hits the option-implied distributions *exactly in the limit*, using an elegant mechanism: the model trains against an adversary that tries to make money trading inside the simulation. When no neural trading strategy can profit, the simulation is arbitrage-free — the training loss literally is the fundamental theorem of asset pricing. And uniquely among generative models, ours ships with a certificate: a computable bound stating "prices computed from this model are wrong by at most this much."

§2 • PROBLEM STATEMENT

The constrained bridge problem

Fix maturities $t_1 < t_2 < \dots < t_K$ at which liquid options trade. By the Breeden–Litzenberger identity, the call-price curve at maturity t_k determines a risk-neutral marginal law μ_k for the asset price X_{t_k} . The object we must learn is a law P on path space satisfying two constraint families:

- **Martingale constraint** $\mathcal{M}$: $\mathbb{E}_P[X_t - X_s \mid \mathcal{F}_s] = 0$ for all $s < t$ — no arbitrage under the pricing measure.
- **Marginal constraints** $\mathcal{P}$: $X_{t_k} \sim \mu_k$ for each k — exact consistency with the option surface.

Among all such laws we select the canonical one: the minimizer of relative entropy $KL(P \parallel R)$ against a reference diffusion R — the *entropic martingale optimal transport* (entropic MOT) solution, the martingale analogue of the Schrödinger bridge. The critical structural observation, which the entire algorithm rests on, is that **both constraint families are affine in the measure P** . The marginal constraints are the standard ones from optimal transport; the martingale constraint, though it involves conditional expectations, is a family of linear equality constraints $\mathbb{E}_P[h(X_{\leq s})(X_t - X_s)] = 0$ indexed by bounded adapted test functions h . Entropy minimization over an intersection of affine sets is exactly the setting where alternating *I-projections* — the generalization of Sinkhorn's algorithm — carry a classical convergence theory.

A formulation note (v1.1 correction). MFM is formulated in *discrete time* — the maturities plus a simulation grid — where relative entropy is finite and the projection theory below applies verbatim. This is not a convenience but a necessity: in continuous time, martingale laws with different volatility functions are mutually singular, so $KL(P \parallel R)$ is identically infinite and the naive continuous-time entropic objective is ill-posed. The literature-consistent continuous-time formulations replace KL with *specific relative entropy* ([Backhoff-Zhang 2026](#)) or penalize drift entropy plus a quadratic volatility cost (the [Schrödinger–Bass bridge](#)); the paper discusses the continuous-time limit in those terms. The first draft of this proposal did not make this distinction — the literature review caught it.

Why is this unsolved? Classical MOT solvers are linear programs on discretized grids: they die beyond two periods or two assets. A Sinkhorn lineage goes further than the v1.0 draft credited: Guyon's [dispersion-constrained martingale Schrödinger problems](#) (Risk 2020; Finance & Stochastics 2024) exactly calibrate joint SPX/VIX smiles by an extended Sinkhorn algorithm, De March–Henry-Labordère run entropic-MOT Sinkhorn for surfaces, and the [martingale Sinkhorn algorithm](#) computes Bass martingales in any dimension with a linear-rate fixed point (Acciaio–Marini–Pammer). All of these are grid-based, nonparametric-on-a-mesh, and low-dimensional. The stochastic-analysis community has likewise established the *existence and structure* theory — Bass martingales, martingale Schrödinger bridges, calibration bridges ([Oct 2025](#), [2026](#)) — by PDE methods with no learning component. On the ML side, flow matching and stochastic interpolants handle marginal constraints beautifully but have never been extended to conditional-mean constraints. The precise claim is therefore: MFM is the first *neural, simulation-free, path-space* algorithm for this problem, with statistical guarantees — the relationship it bears to grid martingale-Sinkhorn is the one neural OT bears to classical Sinkhorn. The grid methods become its exact low-dimensional benchmarks.

§3 · THE ALGORITHM

Alternating projections, made neural

MFM is an alternating KL-projection scheme in path space, with each projection implemented by a scalable neural estimator rather than a grid.

MARTINGALE FLOW MATCHING – TRAINING LOOP

1. **Extract targets.** From the option surface, estimate marginals $\mu_1, \dots, \mu_K$ via Breeden–Litzenberger; project them onto the convex-order cone to remove estimation noise (a small isotonic-type convex program), guaranteeing feasibility by Theorem 1.
2. **Marginal step (projection onto $\mathcal{P}$).** Given the current model, apply a bridge-matching / flow-matching correction that transports each intermediate marginal onto its target μ_k . Simulation-free: the loss is a conditional-expectation regression on interpolant samples, as in standard flow matching.
3. **Martingale step (projection onto $\mathcal{M}$).** Two coupled components:
 - *Adversarial defect estimation.* A discriminator network parameterizes a bounded predictable trading strategy H_φ and is trained to maximize its expected trading gain inside the simulation. By Lemma 3 the maximal gain is the martingale defect $\mathfrak{d}(\mathbf{P})$ — the generator minimizes it.
 - *Drift removal.* The generator's drift is eliminated by a Markovian (mimicking) projection: a regression $\sigma_\theta^2(x, t) = \mathbb{E}[\sigma_t^2 \mid X_t = x]$ that replaces the current process with the local-volatility martingale sharing its one-dimensional marginals (Theorem 4).
4. **Certificate.** At convergence, report the measured marginal errors $W_1(\hat{\mu}_k, \mu_k)$ and the residual defect $\mathfrak{d}(\hat{\mathbf{P}})$. In the single-asset case, Theorem 5 converts these two numbers into an explicit pricing-error bound for Lipschitz path-dependent payoffs.

The output is a neural local-volatility diffusion $dX_t = \sigma_\theta(X_t, t) dW_t$ (or its conditional extension for dynamic simulation), sampled at generation time like any diffusion model — but drift-free by construction and calibrated to the surface. The adversarial component deserves emphasis: it is not a GAN bolted on for realism. The discriminator class is the space of trading strategies, its objective is expected P&L, and its equilibrium value is zero precisely when the fundamental theorem of asset pricing holds inside the model. The economics and the training loss are the same object.

The theorems, with verification status

WHAT "VERIFIED" MEANS HERE

Each result below was audited two ways: its proof strategy was checked against the published mathematics it must rest on, and its novelty was checked against a literature sweep through August 2026. Three status levels are used honestly. **VERIFIED · CLASSICAL** means the result follows by assembling published theorems — the proof is citation plus routine adaptation. **STRATEGY SOUND** means the proof route is standard machinery applied carefully, with the specific technical conditions named below; no step is speculative, but the write-up is real work. **OPEN · RISK ITEM** means genuinely new mathematics is required; the plausibility case and the de-risking plan are stated. A paper claiming everything in the third category is already proven would be dishonest; a paper with nothing in that category would not be a contribution.

THEOREM 1 Feasibility

VERIFIED · CLASSICAL

If the call surface at maturities $t_1, \dots, t_K$ is free of static arbitrage, then the implied marginals exist and increase in convex order, $\mu_1 \preceq_{\text{cx}} \mu_2 \preceq_{\text{cx}} \dots \preceq_{\text{cx}} \mu_K$, and the constraint set $\mathcal{M} \cap \mathcal{P}$ is non-empty. Under compact support of the marginals and a reference diffusion R with bounded, non-degenerate volatility, $\mathcal{M} \cap \mathcal{P}$ contains an element of finite relative entropy with respect to R , so the entropic problem is well-posed.

RESTS ON Strassen (1965) and Kellerer (1972): marginals in convex order are exactly the marginals of a (Markov) martingale. Finite-entropy elements: the stretched-Brownian-motion / Bass-martingale constructions of Backhoff-Veraguas, Beiglböck, Huesmann & Källblad (2020), extended along the multi-maturity chain.

VERIFICATION NOTE The finance structure does real work here: static no-arbitrage of the input surface is *equivalent* to the convex-order condition that Kellerer's theorem needs. Feasibility is not an assumption — it is implied by the data being legitimate. The convex-order projection in Algorithm step 1 makes this robust to estimation noise in real surfaces.

THEOREM 2 Convergence of the alternating scheme

STRATEGY SOUND

The sets $\mathcal{M}$ and $\mathcal{P}$ are affine and closed in the topology of the entropy functional. The alternating I-projection sequence — project the current measure onto $\mathcal{M}$, then onto $\mathcal{P}$, and repeat — converges in relative entropy to the I-projection of the reference R onto $\mathcal{M} \cap \mathcal{P}$, i.e., to the entropic MOT solution.

RESTS ON Csiszár's I-projection theory (1975): alternating entropy projections onto closed affine constraint sets converge to the projection onto the intersection, given a feasible point of finite entropy — which Theorem 1 supplies.

VERIFICATION NOTE Two technical points were checked and must be handled explicitly in the write-up. (i) The martingale set is cut out by *countably many* linear constraints, not finitely many; closedness and lower semicontinuity in the appropriate topology need the standard but careful argument. (ii) The proof must stay primal. Duality-based arguments are a known trap in this area — dual attainment can genuinely fail for martingale transport — so the paper's proof route deliberately avoids the dual, which is why the classical I-projection machinery is the right tool and not merely a convenient one.

V1.1 UPDATE The literature review found that the one-period scalar core of this theorem is now proven: [Chen, Conforti, Ren & Wang](#) (Math. of OR) establish duality without assumed potentials, no primal–dual gap, and exponential Sinkhorn convergence for entropic MOT on $\mathbb{R}$. This is good news twice over — the foundation is firmer than assumed, and the theorem's novel content is sharpened to what is genuinely missing: the **multi-period path-space extension** and, more importantly, an **inexact-projection analysis** — convergence of the alternating scheme when each projection is executed by a neural regression with statistical and approximation error ε , reaching an $O(\varepsilon)$ -neighborhood of the bridge. No error-propagation analysis of this kind exists for neural Sinkhorn-type schemes (DSB and DSBM included), which makes this component a standalone ML contribution.

LEMMA 3 No-arbitrage as a trainable adversarial loss

VERIFIED • ELEMENTARY

A path measure P is a martingale measure if and only if the optimal expected trading gain over bounded predictable strategies is zero: $\mathfrak{d}(P) := \sup_{|H| \leq 1} \mathbb{E}_P[\sum_t H_t \cdot (X_{t+1} - X_t)] = 0$. The functional $\mathfrak{d}$ is an integral probability metric–type defect over the class of predictable integrands, and restricting H to a neural discriminator class yields a consistent lower bound that is tight as the class grows dense.

RESTS ON The definition of conditional expectation, plus a standard density argument for the neural class. The forward direction is immediate; the converse takes $H_t = \text{sign}(\mathbb{E}[X_{t+1} - X_t \mid \mathcal{F}_t])$.

VERIFICATION NOTE Elementary, and deliberately so — this is the load-bearing bridge between the economics and the optimization. It converts the fundamental theorem of asset pricing into a minimax training objective with a quantitative residual (the measured defect feeds Theorem 5's certificate), and it is the reason the discriminator has an economic interpretation rather than being a generic critic.

THEOREM 4 Drift removal by Markovian projection

VERIFIED • CLASSICAL

Let X be an Itô process with bounded, non-degenerate coefficients. There exists a local-volatility martingale with the same one-dimensional marginal laws, with $\sigma^2(x,t) = \mathbb{E}[\sigma_t^2 \mid X_t = x]$.

Consequently the martingale-projection step is implementable as a conditional-expectation regression, and the empirical regressor is consistent under standard statistical-learning conditions.

RESTS ON Gyöngy's mimicking theorem (1986); consistency of least-squares conditional-expectation estimators is textbook nonparametric regression.

VERIFICATION NOTE The boundedness and non-degeneracy conditions become explicit assumptions on the neural parameterization (clipped, positively-bounded σ_θ) — cheap to enforce, and they simultaneously serve Theorem 1's finite-entropy requirement. One honest scope note: for non-Markovian ground truths (rough volatility), what MFM recovers is the Markovian mimicry with the correct marginals and martingale property — exactly the right object for pricing European and many path-dependent claims, and the limitation is stated, not hidden.

THEOREM 5 Pricing-error certificate (single asset)

SUPERSEDED BY PART II (T-C): INTERVAL FORM

Target statement, dimension one: for any payoff F that is L -Lipschitz with respect to the adapted Wasserstein distance, the pricing error of the learned model $\hat{P}$ against the true entropic bridge P^ satisfies $|\mathbb{E}_{\hat{P}}F - \mathbb{E}_{P^*}F| \leq C \cdot L \cdot (\sum_k W_1(\hat{\mu}_k, \mu_k) + \mathfrak{d}(\hat{P}))$ — pricing error is controlled by the two quantities the training loop measures.*

RESTS ON Lipschitz payoffs and adapted Wasserstein continuity (standard); one-dimensional stability of martingale optimal transport, established by Backhoff-Veraguas & Pammer (2022). The genuinely new step is bounding the adapted distance to the bridge by marginal error plus martingale defect.

VERIFICATION NOTE – INCLUDING A REPAIRED ERROR The first draft of this proposal stated this theorem in general dimension. That version is **false**: Brücknerhoff & Juillet (2022) proved martingale transport is unstable in dimension ≥ 2 , so no such certificate can hold there in general. The repaired paper states the theorem for a single underlying across maturities — the economically central case, covering the index-options market — cites the instability result to scope it, and treats multi-asset settings empirically. The impossibility citation is a feature: it tells the reviewer the boundary of the possible was checked, and it explains the paper's theory/empirics split as forced by mathematics rather than convenience.

V1.1 UPGRADE – THE MISSING COMPONENT EXISTS The literature review found both halves of the bound already published. [Blanchet, Wiesel, Zhang & Zhang](#) characterize the adapted-Wasserstein *projection distance onto martingale couplings* explicitly, with parametric convergence rates — the "defect controls distance-to-martingale-class" component, in the one-step case. [Bartl & Wiesel](#) prove multiperiod pricing, hedging, and stopping values are *Lipschitz in adapted Wasserstein distance* — the "distance controls pricing error" component, which also defines the certificate's payoff class citably. What remains new is one composition lemma: extending the projection bound from one step to the multi-maturity chain and composing it with 1D entropic-bridge stability. Status accordingly upgraded from *open* to *strategy sound*.

DE-RISKING GATE Unchanged in substance, smaller in scope: prove the multi-step composition in the discrete-time, finitely-supported setting (backward induction on maturities). Two-week gate before implementation; the target is now a single lemma rather than the full bound. A further statistical upgrade is available from [adapted empirical measure rates](#): composing them with the certificate yields a *finite-sample* pricing guarantee — the form a model-validation function actually consumes.

Verification ledger

CLAIM	RESTS ON	STATUS
Arbitrage-free surface $\Rightarrow$ convex-order marginals $\Rightarrow$ feasibility	Strassen 1965; Kellerer 1972; Breeden–Litzenberger	VERIFIED
Finite-entropy feasible point exists	Bass-martingale constructions, Backhoff-Veraguas et al. 2020	VERIFIED
Both constraint sets affine in the path measure	Direct linearity check (conditional-mean constraints are linear in P)	VERIFIED
Alternating projections converge to entropic MOT	Csiszár 1975; one-period case proven by Chen–Conforti–Ren–Wang 2024–26; novelty = multi-period + inexact neural projections	STRATEGY SOUND
Entropy functional well-posed (discrete time; continuous time via specific relative entropy)	Backhoff–Zhang 2026; Schrödinger–Bass 2026 — v1.0's continuous-time KL was ill-posed, repaired in v1.1	REPAIRED
Martingale $\Leftrightarrow$ zero optimal trading gain (adversarial loss)	Elementary conditional-expectation duality	VERIFIED
Drift removal preserves marginals, yields martingale	Gyöngy 1986	VERIFIED
Pricing certificate, $d = 1$	Blanchet–Wiesel–Zhang–Zhang 2024 (martingale projection in AW) + Bartl–Wiesel 2022 (AW-Lipschitz pricing) + Backhoff–Pammer 2022; one new composition lemma	STRATEGY SOUND
Pricing certificate, $d \geq 2$	Impossible unsmoothed — Brücknerhoff–Juillet 2022; Gaussian-smoothed version conjectured via the regularized Kellerer theorem (Pammer–Robinson–Schachermayer)	EXCLUDED / FRONTIER

§5 · MACHINE-LEARNING NOVELTY

What is new for ICLR, independent of finance

A finance-flavored paper survives the ICLR main track only if its contribution stands without the application. Four contributions do:

- **A new constraint class for flow matching.** The constrained-generation literature handles marginal constraints (Schrödinger bridges, bridge matching), support constraints (reflected and mirror diffusion), and symmetry constraints (equivariant flows). Conditional-mean constraints on the *process* — "the generated dynamics must be a martingale" — are a structurally different affine family that no existing method addresses. MFM extends the stochastic-interpolant toolkit to this class, with convergence theory, and the technique plausibly transfers to any conditional-moment constraint (fairness-in-dynamics, conservation laws in learned simulators).
- **A discriminator with a theorem attached.** Adversarial training is usually a heuristic for realism. Here the discriminator class (trading strategies), its objective (expected P&L), and its equilibrium (zero gain $\Leftrightarrow$ no arbitrage $\Leftrightarrow$ martingale) are a single mathematical object, and its residual value is not just a loss — it is an input to the model's error certificate. Generative models whose adversarial losses carry quantitative semantics are rare.
- **A certificate, not a score.** Generative models are evaluated by FID-style proxies. MFM's evaluation is a provable bound on downstream task error (pricing), computed from training-time quantities. This reframes "how good is my generative model" as a question with a certified answer — a template the ML community can reuse wherever generated data feeds a Lipschitz downstream functional.
- **Scaling a problem the mathematicians just opened.** The existence and structure theory for martingale Schrödinger bridges is 2025–26 mathematics solved by grids and PDEs in dimension one or two. MFM is the simulation-free, regression-based algorithm that scales it — the same relationship neural OT papers had to Sinkhorn, at the moment that relationship was most valuable.

Positioning against the literature (audited August 2026)

PRIOR WORK	WHAT IT DOES	WHAT MFM ADDS
Martingale Schrödinger bridges for SV calibration (2025)	Grid/PDE solution of the calibration bridge, low dimension	Simulation-free neural losses; scales in dimension; learning-theoretic guarantees
Schrödinger–Bass bridge (2026)	Structure theory: optimal process is a stretched Schrödinger bridge	An algorithm; their structure theorem becomes our initialization and sanity check
Arbitrage-free neural-SDE market models (2021–23)	Parametric neural SDEs, no-arbitrage by construction	Exact marginal consistency with the surface (not parametric-family-limited); optimality (entropic selection); certificate
Latent flow matching for IV surfaces (2026); SANOS / DYSANOS	Generative surfaces, soft arbitrage penalties or surface parameterization	Path measures, not surfaces; hard constraints in the limit with convergence proof; dynamics under the pricing measure

PRIOR WORK	WHAT IT DOES	WHAT MFM ADDS
Signature/GAN market generators (2020–24)	Realistic path statistics under the physical measure	The pricing-measure object: martingality and surface consistency, which realism-focused generators do not address
Guyon's dispersion-constrained martingale Schrödinger problems (2020–24) <i>[added in v1.1]</i>	Exact joint SPX/VIX calibration via extended Sinkhorn, on grids	Neural path-space scaling; Guyon's dispersion constraints slot in as a third affine projection — a flagship experiment with an exact benchmark
Martingale Sinkhorn + Bass local-vol fixed point (Conze–Henry–Labordère; Acciaio–Marini–Pammer) <i>[added in v1.1]</i>	Linear-rate grid algorithms for Bass martingales, any dimension	Simulation-free neural estimators; multi-maturity chains; statistical guarantees; also candidate initializers for MFM
Constrained flow matching: physics-constrained , PolyFlow , mirror FM (2025–26) <i>[added in v1.1]</i>	Hard samplewise/state constraints via corrections, architecture, or mirror maps	A martingale constraint cannot be enforced samplewise — it couples the law across the filtration; conditional-moment constraints remain an open class these methods do not reach

Novelty was checked twice: a targeted sweep in the original draft, then a full 37-paper literature review (August 2026) covering the martingale-transport, adapted-Wasserstein, entropic-Sinkhorn, and constrained-flow-matching pillars. The review confirmed no work combines flow matching or stochastic interpolants with martingale constraints, while surfacing the Sinkhorn lineage rows above and upgrading two theorems (§4). One more asset it produced: [Blanchet et al.'s martingale-coupling hypothesis test](#) — already applied in that paper to detect arbitrage in neural generative pricing models — becomes MFM's *external, published evaluation metric*: every baseline is scored by the field's own arbitrage test, which MFM alone should pass by construction. The sweep should be repeated immediately before submission; this area moved visibly between June and August 2026 alone.

§6 • EXPERIMENTAL PROGRAM

Detailed analysis: experiments, baselines, metrics

E1 — Exact recovery on synthetic ground truth

Generate surfaces from Heston and rough Bergomi; run MFM on the surfaces alone. Metrics: marginal Wasserstein error at each maturity, measured martingale defect, and repricing error on vanillas (should be near-zero by construction) and on exotics (barriers, Asians, lookbacks) against Monte Carlo ground truth. For rough Bergomi, report the Markovian-mimicry gap explicitly — this is the honest version of the experiment, and it doubles as an empirical measurement of what Theorem 4's scope note costs in practice.

E2 — Tightness against exact solvers, low dimension

In one and two periods where LP-based MOT is exactly solvable, verify MFM converges to the entropic solution and, as the entropic regularization is annealed, approaches the true MOT price bounds. This is the experiment that convinces the mathematically literate reviewer: the neural method is checked against a solver with no learning in it.

E3 — SPX calibration and exotic pricing

Calibrate to full SPX surfaces across regimes (calm 2017, COVID 2020, hiking-cycle 2022). Price cliquets and barriers; compare against local vol (Dupire), the neural-SDE market model baseline, and the grid-based martingale Schrödinger bridge where dimension permits. Report the Theorem 5 certificate alongside each price — the headline table of the paper is prices *with certified error bars*.

E4 — Scaling where grids cannot go

Multi-asset baskets ($d = 5\text{--}50$) with marginals from index-component surfaces. No exact solver exists here; the claims are empirical: defect and marginal errors at scale, wall-clock versus grid methods (which are infeasible), and sample-quality comparisons against penalty-based generators. Scoped honestly per the $d \geq 2$ instability: measured errors, no certificates.

E5 — Downstream: does arbitrage-freeness matter?

Train identical deep-hedging agents on (a) MFM, (b) a soft-penalty generator, (c) historical bootstrap. Evaluate on held-out real data. Hypothesis: agents trained on (b) partially learn to harvest simulator arbitrage and degrade out-of-simulator; the gap is the economic value of the constraint. This experiment converts the paper's mathematics into a number a practitioner cares about, and it is the section that makes the "financial infrastructure" claim non-rhetorical.

Risk register

RISK	SEVERITY	MITIGATION
Theorem 5's new bound resists proof	High — ceiling, not floor	Two-week discrete-case gate before implementation; paper stands on T1–T4 + empirics if it fails
Entropic bias: bridge $\neq$ market's actual measure	Medium	ϵ -annealing toward true MOT; report price bounds, not point prices, where it matters; E2 quantifies the bias
Noisy real surfaces break convex order	Medium	Convex-order projection preprocessing (Algorithm step 1) — restores Theorem 1's hypothesis by construction
Adversarial step trains unstably	Medium	Defect is an IPM with bounded critics — the stable end of adversarial training; drift-removal regression does most of the work, discriminator is a residual check

RISK	SEVERITY	MITIGATION
Scooped between now and submission	Low-medium	Re-run the literature sweep at submission; the specific combination (FM + martingale constraint + certificate) has three defensible components, unlikely all fall at once

Anticipated reviewer objections

- *"Isn't this just Sinkhorn for MOT?"* — Sinkhorn-type grid schemes exist and are cited; they are exponential in dimension and offer no learning-theoretic account. MFM's projections are simulation-free regressions and an IPM game, the estimators are neural, and the certificate is new. The relationship is the one neural OT bears to Sinkhorn — and that literature is core ICLR.
- *"The entropic solution is not the market's measure."* — Correct, and no method that selects one martingale measure can claim otherwise; the market's measure is underdetermined by the surface. MFM makes the selection principled (maximum entropy relative to a reference), controllable (annealing recovers bounds), and certified (Theorem 5 bounds the error to the *bridge*; E2/E3 measure the rest).
- *"Why hard constraints instead of penalties, which the 2026 surface papers use?"* — Penalties leave residual arbitrage with no convergence theory; a hedging engine will find the residual (E5 tests exactly this). Projections come with Theorem 2 and an exactness limit. When the constraint is the fundamental theorem of asset pricing, "mostly satisfied" is not a category the application accepts.
- *"Only $d = 1$ theory."* — By necessity, not choice: the certificate is provably impossible in general for $d \geq 2$ (Brücknerhoff-Juillet), and the paper is the first in the ML literature to import that boundary. Scoped theory plus scaled empirics is the honest frontier.

Execution plan

Twelve weeks of core work: weeks 1–2, the Theorem 5 discrete-case gate and the Theorem 2 write-up in parallel; weeks 3–6, implementation and E1–E2 (the exact-solver comparisons are the debugging harness, not just an experiment); weeks 7–10, E3–E5; weeks 11–12, the multi-asset scaling study and paper assembly. The theory gate comes first deliberately: it is the cheapest experiment with the highest information, and every downstream claim inherits its outcome. Target the next ICLR cycle if the gate clears immediately; otherwise the identical paper targets the following ICML/NeurIPS window without loss — the lane is protected less by speed than by the fact that it sits between two communities, neither of which is currently equipped to cross it.

§7 · THE INFRASTRUCTURE CLAIM

Why this is a foundation, not a paper-sized trick

The claim that this founds something should be defended, not asserted. Three properties qualify it:

It sits at the correct abstraction layer. Every derivatives-facing system — pricing, XVA, hedging, margin, stress testing — consumes scenarios under a pricing measure. Today each system embeds its own model of that measure. A generative object that is provably martingale, exactly surface-consistent, optimally selected, and certificate-carrying is the shared substrate those systems can consume: one calibrated simulator, many consumers, each inheriting the certificate. That is what infrastructure means — the layer beneath the applications, with a contract.

Its guarantees are the kind regulation can consume. Model risk frameworks (SR 11-7 and successors) demand quantified model error. A generative model whose output ships with a provable pricing-error bound is the first of its kind that a validation function can sign off on by checking two measured numbers against a theorem, rather than by judgment. The certificate is not a paper ornament; it is the compliance interface.

It opens a program, not a result. Immediate successors, each a paper: conditional MFM as a dynamic market simulator (condition on today's state, regenerate the bridge daily); the *smoothed* multi-asset certificate — the literature review found that Gaussian regularization restores Kellerer's theorem in arbitrary dimension ([Pammer–Robinson–Schachermayer](#)) and simultaneously fixes the statistical rates, so a bandwidth-parameterized $d \geq 2$ certificate is a concrete conjecture with a fast falsification test (does the Brücknerhoff–Juillet counterexample survive smoothing?); a *Bass-architectural* variant where the generator is the gradient of an input-convex network composed with a Brownian bridge — martingale by architecture per the [Schrödinger–Bass structure theorem](#), trained by the linear-rate fixed point instead of an adversarial game; dispersion-constrained MFM for joint SPX/VIX calibration at scale (Guyon's constraints as a third projection); conditional-moment-constrained flow matching outside finance — conservation laws in learned physics simulators are the same affine constraint class; and the certificate template for any generative model feeding a Lipschitz downstream task. A foundation is measured by what can be built on it without re-deriving it. This qualifies.

PART II · V2.0 – THE DEVELOPED THEOREMS, ADVERSARIALLY VERIFIED

Six developments, refereed hostilely before submission

VERIFICATION PROTOCOL & WHY THE ERRORS ARE LISTED

Each development below was drafted in full mathematical detail and then attacked by a 13-agent verification: a max-effort *hostile referee* per item (instructed only to refute, checking every inequality and constructing explicit numerical counterexamples), a *literature auditor* per item (confirming every citation says what we use it for, hunting prior art), and a *completeness critic* over the whole suite (cross-item consistency, composed constants, shared hidden assumptions). Four of six items were refuted *as first stated* and repaired into stronger final forms. The refutations are recorded openly: a suite that survived hostile refereeing unchanged would be evidence the refereeing was toothless, and every recorded counterexample is now either a scope note or a publishable result in its own right. Status vocabulary: **proven-as-stated** = the final statement's proof was verified step-by-step after repair; **provable** = route standard and checked, write-up remains; **assumption-explicit** = correct under named, checkable assumptions; **conjecture** = labeled, used by nothing.

T-A Defect–Distance Equivalence

PROVEN AS STATED ($\mathbb{R}^d$)

For every integrable path law P on $(\mathbb{R}^d)^{T+1}$: $\mathfrak{d}(P) \leq 2 \cdot \text{AW}_1(P, \mathcal{M})$ and $\text{AW}_1(P, \mathcal{M}) \leq T \cdot \mathfrak{d}(P)$ — the adversarial no-arbitrage loss is, up to constants $(2, T)$, the adapted-Wasserstein distance to the arbitrage-free class, and the constant 2 is sharp (attained by the path $(0, 1, 0)$).

WHAT BROKE → FIX The grid-supported form is *false with unbounded violation ratio*: the referee's triangle counterexample (three affinely independent grid points force every grid-supported martingale to be constant) drives the ratio to infinity. Restated on $\mathbb{R}^d$, every step of both proofs was then verified line-by-line — the mean-shift construction with its adaptedly-invertible (hence bicausal) coupling, and the anticausality argument for the lower bound. The triangle example is kept as a published warning: grid-side conversions must use a grid-geometry constant, never this theorem. One-step antecedent: [Blanchet–Wiesel–Zhang–Zhang](#); new: multi-step, two-sided, sharp constant, and the identification of the FTAP-GAN loss with a metric.

T-B Inexact cyclic I-projections (budget form)

PROVABLE • WEAKENED CORRECTLY

The cyclic scheme (one projection per marginal plus the martingale set, started at R , on time-indexed grids $K_t \subseteq \text{relint conv}(K_{\{t+1\}})$) with per-step inexactness ε_n and $\sum \sqrt{\varepsilon_n} < \infty$ converges to some $P_\infty \in \mathcal{M} \cap \mathcal{P}$ with $\text{KL}(P^* \| P_\infty) \leq C \cdot \sum (\varepsilon_n + \sqrt{\varepsilon_n})$; convergence to the bridge P^* itself is recovered under dual-coordinate inexactness or a Dykstra correction.

WHAT BROKE → FIX Both original conclusions were refuted by explicit numerical counterexamples: a single inexact step can land on a different feasible point where all exact projections stall forever, and constant- ε errors drift *tangentially along the feasible set* with no restoring force. The corrected budget form is what is actually true — and the two counterexamples are themselves new recorded results explaining why naive inexact IPF loses the bridge. The literature pass also found [exponential convergence of general IPF including martingale constraints](#) (2025), so the exact-scheme convergence is no longer the novelty; the inexact/budget theory with neural oracles and a computable dual-residual surrogate is — an error-propagation theory that DSB/DSBM also lack. The original Slater condition was *unsatisfiable* on a time-constant grid (extreme points absorb martingales); the time-indexed-grid setting repairs it.

T-C Robust-pricing certificate — interval form

PROVABLE $D=1$ • ASSUMPTION-EXPLICIT $D \geq 2$

$\text{dist}(\mathbb{E}_{\hat{P}} F, [V_{\min}(\mu), V_{\max}(\mu)]) \leq L \cdot T \cdot \delta(\hat{P}) + L \cdot \omega(\sum_k W_1(\hat{\mu}_k, \mu_k) + T \cdot \delta(\hat{P}))$ — the model's price provably lies within a computable distance of the arbitrage-free price interval, with the explicit nonlinear modulus ω from quantitative 1D stability.

WHAT BROKE → FIX The v1.1 certificate targeted a single bridge measure — *structurally false*, refuted independently by two referees and my own audit: two exactly-calibrated zero-defect martingales price the same Lipschitz payoff differently (2 vs 5/3 in the verified example), so the right-hand side vanishes while the left does not. Calibrated martingale measures are non-unique — that is the textbook reason MOT price *intervals* are nondegenerate — and the interval is therefore the only correct target. The interval form is also stronger science: it is regulator-consumable, and it certifies the output of *either* training track (the KL track of T-B or the Bass track of T-E), resolving a two-selection-principles inconsistency the completeness critic caught. Also fixed: the linear stability constant (unsupported — replaced by Wiesel's modulus $\omega(r) = \inf_{\delta} [K(\delta)r + \delta]$), the (A3) over-attribution (now a named, numerically checkable assumption), and the undefined entropic temperature in the $d \geq 2$ regime (now a declared reference R_g ; fixed-grid statement, mesh-uniformity explicitly open). Bridge-proximity survives only as a separate corollary requiring the T-B track's KL-optimality gap.

T-D Smoothed multi-asset program

P1-P3 PROVEN (SCOPED) • C1 CONJECTURE

Convex order is preserved under maturity-wise Gaussian smoothing (P1); finite-maturity feasibility in any dimension is Strassen — what fails unsmoothed in $d \geq 2$ and is restored by regularization is Markov/mimicking existence (P2, scoped precisely); and adding an independent Gaussian martingale gives a model calibrated to the smoothed surface with $|\mathbb{E}F(X) - \mathbb{E}F(\tilde{X})| \leq L \cdot \Sigma \mathbb{E}[G_\sharp]$ for static path-Lipschitz payoffs (P3).

WHAT BROKE → FIX P3's adapted-Wasserstein form was refuted by information-masking counterexamples (a tiny signal determining next-period volatility is masked by the noise; a 1-Lipschitz stopping value drops by $\Theta(1)$ while the claimed bound is $O(\sigma)$) — independent noise destroys information, so the coupling is causal but not bicausal. The corrected P3 uses the trivial coupling (unbreakable), explicitly excludes stopping/superhedging transfer, and offers an AW-level corollary only under a kernel-Lipschitz assumption via $\text{AW} \leq C\sqrt{W_1}$. Conjecture C1 (smoothed $d \geq 2$ stability, uniform in temperature) stays labeled, used by nothing, with its falsification test on the Brücknerhoff–Juillet marginals and competing regularization routes cited.

T-E Tower-property (Bass) architecture

PROVABLE • CONSTRUCTION COMPLETED

$X_t = \mathbb{E}[g_\theta(W_1) \mid W_t]$ with a learnable initial law $W_0 \sim \alpha_\theta$ (the Bass measure) and $g_\theta = \nabla(\text{ICNN})$ is an exact martingale by the tower property — martingality by architecture, not by loss — with multi-period chaining via the inverse-heat junction link and the adversarial defect demoted to a quantified audit: $\mathbb{E}[\text{defect}] \leq C \cdot T \cdot \sqrt{d} \cdot \text{Lip}(g_\theta) \cdot N^{-1/2} + \text{junction tolerances, feeding T-C.}$

WHAT BROKE → FIX Three missing components, each exhibited by counterexample: without α_θ the architecture fits only Dirac initial laws (real-analytic heat smoothing can never produce atoms); naive multi-period chaining is not a martingale (cubic-map junction defect computed); and an arbitrary Gaussian basis breaks own-filtration martingality (fBM counterexample) — the basis must have independent increments. The exact-fit claim is scoped to marginals with densities, which makes the architecture's natural habitat the *smoothed surface of T-D* — two independently verified developments that fit together. Novelty narrowed honestly by the literature pass: the construction is Conze–Henry-Labordère (2021, non-neural), and neural Schrödinger–Bass generators now exist ([LightSBB-M](#), [SBBTS](#), 2026); the delta is the ICNN terminal map + tower-exactness as an ML architecture principle + the audited-defect-to-certificate pipeline. Its trained limit is the Bass/MBB martingale — openly the second selection principle, certified by T-C's interval form.

T-F Statistical certificate (finite-sample)

PROVABLE • ALL ERROR TERMS NOW PRESENT

With probability $1 - \delta$, the distance from $\mathbb{E}_P F$ to the arbitrage-free interval is bounded by the T-C terms plus an explicit statistical budget: the neural-class approximation error $\varepsilon_{\text{approx}}$, the Rademacher term of the gain class, Monte-Carlo defect estimation with its own budget m of fresh model samples, and per-marginal empirical W_1 rates — with the failure probability split across the $K+1$ events.

WHAT BROKE → FIX Three verified errors: the single- P^* target (fixed by T-C); a missing approximation term — the neural class *under*-estimates the true defect and no Rademacher term closes that gap (the known neural-net-distance limitation, with a sign-pattern counterexample) — now an explicit $\varepsilon_{\text{approx}}$ under a stated smoothness assumption on conditional means; and a sampling-model conflation — the defect is an expectation under the *model*, so it is Monte-Carlo estimated from $\hat{P}$ with budget m separate from the data size n , and "n surface observations" (an ill-posed Breeden–Litzenberger inverse problem) is split into its own assumption-laden branch. Must-engage prior art found: [Obłój–Wiesel](#) (robust superhedging estimation; the naive plugin's documented failure is exactly the trap the interval target and explicit budget avoid) and the one-period test of Blanchet et al.

The change ledger: errors solved vs. novelty sharpened

#	FINDING	KIND	RESOLUTION
1	Grid form of the defect–distance equivalence false (unbounded ratio; triangle counterexample)	Error, would fall when tested	$\mathbb{R}^d$ restatement, proven; grid warning kept as a result
2	Inexact projections do not converge to the bridge (two counterexamples: limit capture, tangential drift)	Error, would fall when tested	Budget form; P^* recovered via dual-inexactness or Dykstra
3	Single-measure pricing certificate structurally false (calibrated martingales non-unique; zero-RHS counterexamples)	Error, would fall when tested	Interval certificate — also resolves the two-selection-principles inconsistency

#	FINDING	KIND	RESOLUTION
4	Linear stability constant unsupported; (A3) over-attributed; entropic temperature undefined	Errors of attribution/precision	Nonlinear modulus ω ; named checkable assumption; declared R_e
5	Adapted-Wasserstein smoothing bound false (information-masking counterexamples)	Error, would fall when tested	Static-payoff transfer via trivial coupling; conditional AW corollary
6	Bass architecture incomplete (no initial law; junctions not martingales; basis condition missing)	Errors, would fall when tested	Learnable α_0 ; inverse-heat junctions; independent-increment basis
7	Statistical certificate missing approximation term; sampling models conflated	Errors, would fall when tested	$\varepsilon_{\text{approx}}$ + split observation models + separate MC budget
8	Slater condition unsatisfiable on time-constant grids; restoration lemma silently assumed three times	Errors of formulation	Time-indexed grids; standalone Restoration Lemma
9	Two-sided equivalence with sharp constant; interval form as regulator-consumable certificate; T-D/T-E consonance (smoothed surface as the architecture's habitat); T-B repositioned onto the genuinely open inexact theory; T-E delta stated against Conze–Henry-Labordère, LightSBB-M, SBBTS	Novelty improvements (correct before and after)	Final v2.0 suite

The complete corrected statements, all counterexamples in full, and the nine verifier-surfaced references are in the companion repository's DEVELOPMENTS.md; the papers themselves are in its reading-path tiers. A claim of "none will fall when tested" can only honestly mean: everything labeled proven was verified step-by-step by an adversary that had already broken the earlier version; everything else carries its named assumptions in the statement.

§8 · REFERENCES

Key references

- Strassen (1965), The existence of probability measures with given marginals; Kellerer (1972), Markov-Komposition und eine Anwendung auf Martingale — feasibility.

- Csizsár (1975), I-divergence geometry of probability distributions and minimization problems — alternating-projection convergence.
- Gyöngy (1986), Mimicking the one-dimensional marginal distributions of processes having an Itô differential — drift removal.
- Backhoff-Veraguas, Beiglböck, Huesmann, Källblad (2020), stretched Brownian motion / Bass martingales — finite-entropy feasibility.
- Backhoff-Veraguas & Pammer (2022), stability of martingale optimal transport in dimension one; Brücknerhoff & Juillet (2022), instability in dimension ≥ 2 — the certificate's scope, both directions.
- [Martingale optimal transport and martingale Schrödinger bridges for calibration of stochastic volatility models](#) (2025).
- [A weak transport approach to the Schrödinger–Bass bridge](#) (2026).
- [Arbitrage-free neural-SDE market models](#) (2021).
- [Latent flow matching for arbitrage-aware implied volatility surface generation](#) (2026); [SANOS](#) (2026); [DYSANOS](#) (2026) — the penalty-based state of the art MFM supersedes.
- *Added in v1.1:* Guyon (2020–24), dispersion-constrained martingale Schrödinger problems (Risk; Finance & Stochastics); [Chen, Conforti, Ren, Wang](#), Sinkhorn convergence for entropic MOT; [Blanchet, Wiesel, Zhang, Zhang](#), empirical martingale projections in adapted Wasserstein; [Bartl, Wiesel](#), AW-sensitivity of multiperiod optimization; [the martingale Sinkhorn algorithm](#); [Pammer, Robinson, Schachermayer](#), regularized Kellerer theorem in arbitrary dimension; [Backhoff, Zhang](#), specific relative entropy between continuous martingales; [Backhoff, Bartl, Beiglböck, Wiesel](#), estimating processes in adapted Wasserstein.

Classical references are cited without links to avoid misattributed URLs; arXiv links are given only where verified during the August 2026 literature audit.

Prepared as a research proposal for an ICLR main-track submission. v1.1 (August 2026): revised after a 37-paper literature review — prior art added, the continuous-time entropy formulation repaired, theorems repositioned/upgraded. v2.0 (August 2026): the six new developments (Part II) written out in full and put through a 13-agent adversarial verification — hostile referees with numerical counterexamples, literature auditors, and a completeness critic; four of six were refuted as first stated and repaired into stronger final forms, nine further references were surfaced (now 46 papers in the companion repository), and the certificate was restructured onto its provably correct target, the arbitrage-free price interval. The repository's DEVELOPMENTS.md holds the corrected statements and the complete verification record.